

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/01

Paper 1 Multiple Choice

14 September 2012

50 minutes

Additional Materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **11** printed pages and **1** blank page.



Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 Methane was burned in an incorrectly adjusted burner. The methane was converted into a mixture of carbon dioxide and carbon monoxide in the ratio of 98:2, together with water vapour.

What will be the volume of oxygen consumed when $y \text{ dm}^3$ of methane is burned?

- A** $1.99y \text{ dm}^3$ **C** $0.995y \text{ dm}^3$
B $1.995y \text{ dm}^3$ **D** $0.99y \text{ dm}^3$

- 2 To identify an oxide of nitrogen, 0.10 mol of the oxide was mixed with 10 dm^3 of hydrogen gas and passed over a heated catalyst. At the end of the reaction, 0.4 dm^3 of hydrogen gas remained. The ammonia produced required 125 cm^3 of 1.6 mol dm^{-3} HCl for neutralisation. All gaseous volumes were measured at room temperature and pressure.

What is the formula of the oxide of nitrogen?

- A** NO **C** NO_2
B N_2O **D** N_2O_4

- 3 Which atom has the highest ratio of unpaired electrons to paired electrons in its ground state?

- A** Boron **C** Nitrogen
B Carbon **D** Oxygen

- 4 The table shows the successive ionisation energies for an element **Q**.

	1 st	2 nd	3 rd	4 th
Ionisation energy / kJ mol^{-1}	418	3070	4600	5860

What is the likely formula of the oxide of **Q**?

- A** QO **C** Q_2O_3
B Q_3O_2 **D** Q_2O

- 5 Ions of two isotopes of the transition metal nickel are shown below.



Which one of the following statements is correct?

- A** The ${}^{60}_{28}\text{Ni}^{2+}$ ion has more protons than the ${}^{58}_{28}\text{Ni}^{2+}$ ion.
B The electronic configuration of both Ni^{2+} ions is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$.
C These Ni^{2+} ions have the same number of electrons but a different number of neutrons.

- D The ${}^{60}_{28}\text{Ni}^{2+}$ ion will be deflected more than the ${}^{58}_{28}\text{Ni}^{2+}$ ion under the same magnetic field.

6 Which one of the following **cannot** be explained by hydrogen bonds?

- A CH_3COOH has a M_r of 120 in benzene.
B CH_3CHO has a higher boiling point than C_2H_6 .
 C HF has a higher boiling point than HCl.
 D CH_3COCH_3 is miscible with water.

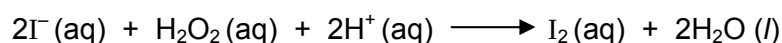
7 Which of the following molecules does **not** have a permanent dipole?

- | | |
|-----------------------------------|------------------------------------|
| A C_2Cl_4 | C CHCl_3 |
| B CF_2Cl_2 | D CF_2CCl_2 |

8 What is the order of increasing H—N—H bond angle in the species NH_2^- , NH_3 and NH_4^+ ?

- A $\text{NH}_4^+ < \text{NH}_3 < \text{NH}_2^-$
B $\text{NH}_2^- < \text{NH}_3 < \text{NH}_4^+$
 C $\text{NH}_3 < \text{NH}_4^+ < \text{NH}_2^-$
 D $\text{NH}_4^+ < \text{NH}_2^- < \text{NH}_3$

9 Hydrogen peroxide reacts with acidified iodide ions liberating iodine.

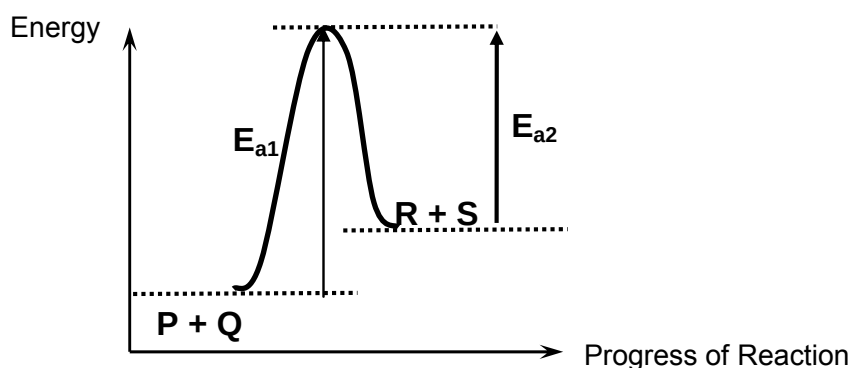


The rate equation for the reaction is found to be $\text{rate} = k [\text{H}_2\text{O}_2][\text{I}^-]$, where k is the rate constant.

Which of the following conclusions **cannot** be drawn from the above information?

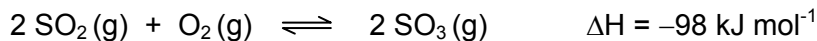
- A** Halving the concentration of iodide and hydrogen peroxide simultaneously halve the reaction rate.
 B The rate of reaction is affected by a change in the iodide concentrations.
 C The overall reaction is second order.
 D The reaction is first order with respect to hydrogen peroxide.

- 10 A certain reversible reaction, $P + Q \rightleftharpoons R + S$, gives the following energy profile diagram.



Which of the following statements is true?

- A $E_{a1} = E_{a2} - \Delta H$
 - B The formation of **R** and **S** is an exothermic reaction.
 - C Temperature of surrounding increases during the formation of **R** and **S**.
 - D** The activation energy of the backward reaction is lower than the activation energy of the forward reaction.
- 11 The key reaction during the Contact process for manufacturing sulfuric acid is as follows:



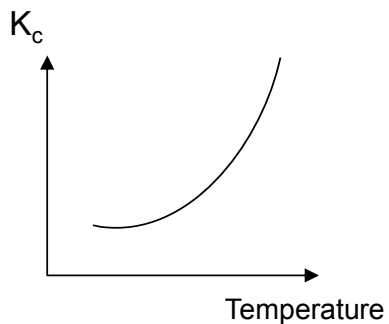
What can be deduced from the following equilibrium?

- A Adding a catalyst increase the yield of sulfur trioxide.
- B Decreasing the temperature increases the rate of formation of sulfur trioxide.
- C Decreasing the temperature will decrease the yield sulfur trioxide.
- D** Increasing the pressure will increase the yield of sulfur trioxide.

12 The equilibrium constant K_c for the reaction



varies with temperature as shown in the diagram below.



What deductions can be made from this information?

- A The yield will increase at higher pressure.
- B** The forward reaction is endothermic.
- C The value of K_c depends on amount of X used.
- D The yield is independent on temperature.

13 Given the following standard enthalpy changes,

$$\Delta H_c^\theta (\text{C}) = -394 \text{ kJ mol}^{-1}$$

$$\Delta H_c^\theta (\text{H}_2) = -286 \text{ kJ mol}^{-1}$$

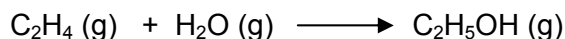
$$\Delta H_f^\theta (\text{C}_2\text{H}_5\text{OH}) = -278 \text{ kJ mol}^{-1}$$

What is the standard enthalpy change of combustion of ethanol, $\text{C}_2\text{H}_5\text{OH}$?

- A -402 kJ mol^{-1}
- B -758 kJ mol^{-1}
- C** $-1368 \text{ kJ mol}^{-1}$
- D $-1924 \text{ kJ mol}^{-1}$

- 14 The use of the Data Booklet is relevant to this question.

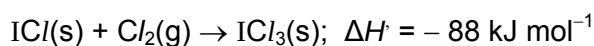
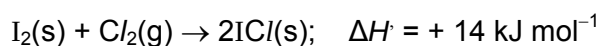
Ethanol is prepared industrially by the reaction of ethene and steam in the presence of a catalyst.



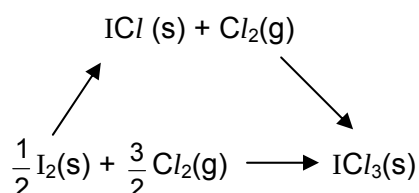
What is the enthalpy change for the reaction?

- A** -50 kJ mol^{-1}
B $+50 \text{ kJ mol}^{-1}$
C -310 kJ mol^{-1}
D $+310 \text{ kJ mol}^{-1}$

15. Iodine trichloride, ICl_3 , is made by reacting iodine with chlorine.



By using the above data and the following energy cycle, what is the enthalpy change of the formation of solid iodine trichloride?



- A** -60 kJ mol^{-1} **B** -74 kJ mol^{-1} **C** -81 kJ mol^{-1} **D** -162 kJ mol^{-1}

16. The table shows the enthalpy change of neutralisation per mole of water formed, ΔH , for various acids and bases.

acid	base	$\Delta H / \text{kJ mol}^{-1}$
hydrochloric acid	sodium hydroxide	-57.0
P	sodium hydroxide	-54.0
hydrochloric acid	Q	-52.0
nitric acid	R	-57.0

What are **P**, **Q** and **R**?

- | | P | Q | R |
|----------|---------------|------------------|---------------------|
| A | ethanoic acid | ammonia | potassium hydroxide |
| B | ethanoic acid | sodium hydroxide | ammonia |
| C | sulfuric acid | ammonia | potassium hydroxide |
| D | sulfuric acid | sodium hydroxide | ammonia |

- 17 A dilute solution of hydrofluoric acid, HF is a weak acid and dissociates as follows:



Given that only 7% of HF is dissociated into ions, calculate the pH of 100 cm³ of 0.1 mol dm⁻³ HF.

- A 1.0
 B 2.0
C 2.2
 D 2.5
- 18 The table gives the concentrations and pH values of the aqueous solutions of two compounds, X and Y. Either compound could be an acid or a base.

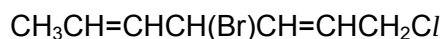
	X	Y
Concentration / mol dm ⁻³	2.0	2.0
pH	6	9

Student P concluded that X is a strong acid.

Student Q concluded that the extent of dissociation is lower in X(aq) than in Y(aq).

Which of the students are correct?

- A P only
B Q only
 C both P and Q
 D neither P nor Q
- 19 Compound X has the following structure:



What is the total number of geometric isomers possible for this molecule?

- A 1
 B 2
 C 3
D 4

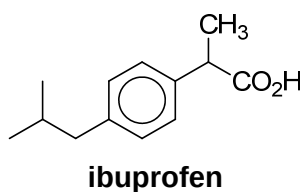
20 An organic compound **X** undergoes the following reactions:

- (i) It decolourises a solution of bromine in tetrachloromethane.
- (ii) It reacts with sodium bromide in concentrated sulfuric acid.
- (iii) It reacts with hot alkali to produce a compound with two alcohol functional groups.

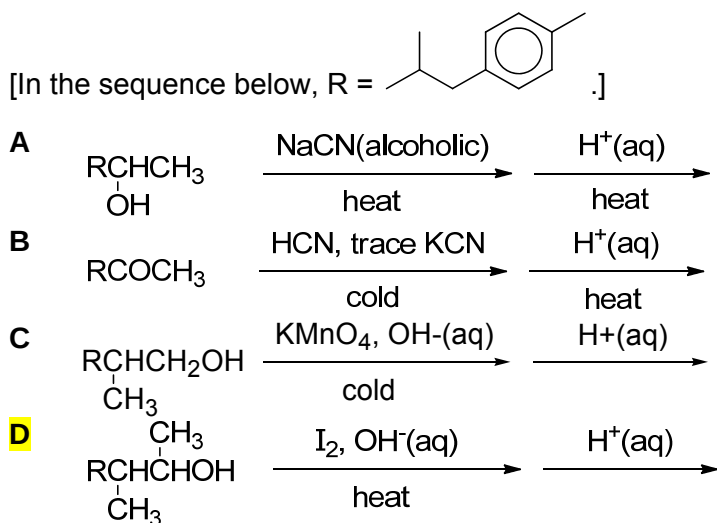
Which compound could be **X**?

- A** $\text{HOH}_2\text{CCH}=\text{CHCH}=\text{CHCH}_2\text{Cl}$
- B** $\text{Cl}_2\text{CHCH}=\text{CHCH}_2\text{COOH}$
- C** $\text{BrCH}_2\text{CH}_2\text{CHClCH}_2\text{COC l}$
- D** $\text{ClCH}_2\text{CH}_2\text{CH}=\text{CHCH}(\text{Cl})\text{CH}_2\text{OH}$

21 Ibuprofen is an anti-inflammatory agent.



Which of the following could be part of a sequence for synthesizing ibuprofen?

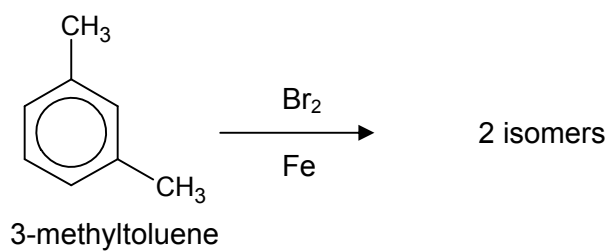


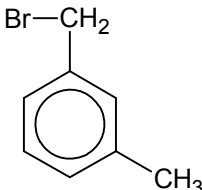
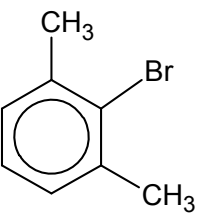
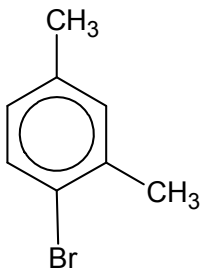
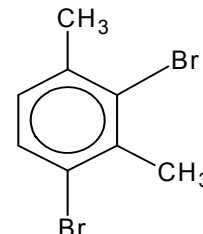
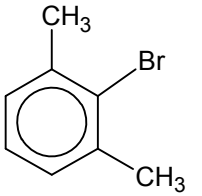
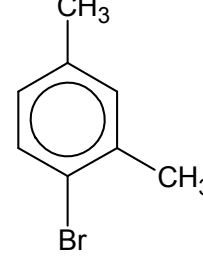
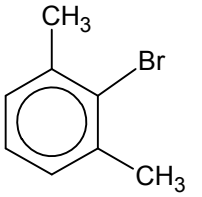
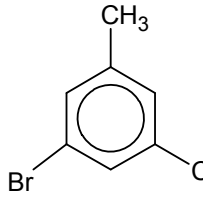
22 In which of the following sequences does the value of K_a decrease continuously?

- A** $\text{C}_2\text{H}_5\text{OH} > \text{CH}_3\text{CO}_2\text{H} > \text{CCl}_3\text{CO}_2\text{H}$
- B** $\text{CCl}_3\text{CO}_2\text{H} > \text{CH}_3\text{CO}_2\text{H} > \text{C}_2\text{H}_5\text{OH}$
- C** $\text{CH}_3\text{CO}_2\text{H} > \text{CCl}_3\text{CO}_2\text{H} > \text{C}_2\text{H}_5\text{OH}$
- D** $\text{C}_2\text{H}_5\text{OH} > \text{CCl}_3\text{CO}_2\text{H} > \text{CH}_3\text{CO}_2\text{H}$

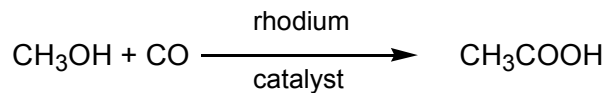
- 23 When 3-methyltoluene is treated with bromine in the presence of iron fillings in the dark, a mixture of two mono-brominated isomers is formed.

What is the structure of these two isomers?

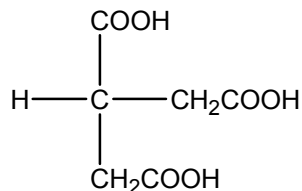


- A**
- 

- B**
- 

- C**
- 

- D**
- 


- 24 Ethanoic acid can be made by direct carbonylation of methanol, in presence of a rhodium catalyst.



Using a similar method, which one of the following compounds can produce:



- A**
- $$\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{COOH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
- B**
- $$\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$$
- C**
- $$\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{COOH} \end{array}$$
- D**
- $$\begin{array}{c} \text{OH} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{COOH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$

- 25 Consecutive elements **P**, **Q**, **R** are in the third period of the Periodic Table. Element **Q** has the highest first ionisation energy and the lowest melting point.

What could be the identities of **P**, **Q** and **R**?

- A** magnesium, aluminium, silicon
B aluminium, silicon, phosphorus
C sodium, magnesium, aluminium
D silicon, phosphorus, sulfur

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

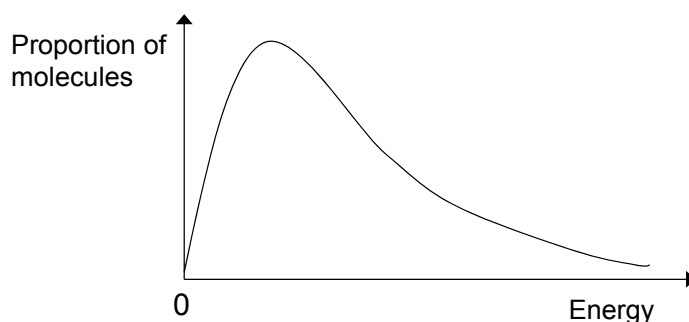
A	B	C	D
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct.

26 Which of the following would result in an ionic compound showing more covalent character?

- 1** The radius of the cation is large rather than small.
- 2** The radius of the anion is large rather than small.
- 3** The charge on the cation is large rather than small.

Answer: C

27 The diagram represents the Boltzmann distribution of molecular energies at a given temperature.



As temperature increases, which statements are correct?

- 1** The maximum of the curve is displaced to the right.
- 2** The proportion of molecules with energies above any given value increases.
- 3** The number of molecules at any given energy increases.

Answer: B

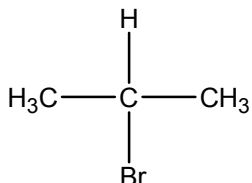
28 Which statements about the effect of a catalyst on a reversible reaction are correct?

- 1 It increases the yield of product in an equilibrium.
- 2 It reduces the activation energy of the forward reaction and reverse reaction.
- 3 It increases the rate constant for both the forward reaction and reverse reaction.

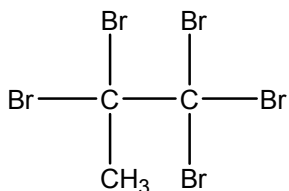
Answer: C

29 Which compound(s) could undergo elimination reaction when treated with hot ethanolic potassium hydroxide?

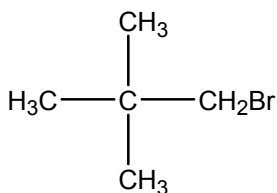
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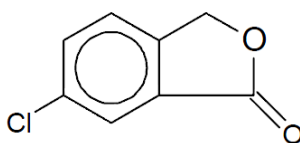


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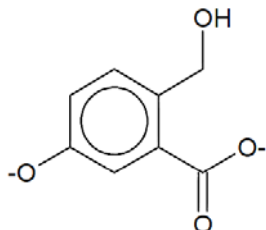


Answer: B

30 Which of the following statements is **true** for the compound below?



- 1 It is a non-planar compound.
- 2 When boiled with aqueous sodium hydroxide, it is converted to



- 3 It will react with 2,4-dinitrophenylhydrazine at room temperature.

Answer: D

1	A	11	D	21	D
2	B	12	B	22	B
3	C	13	C	23	C
4	D	14	A	24	D
5	C	15	C	25	D
6	B	16	A	26	C
7	A	17	C	27	B
8	B	18	B	28	C
9	A	19	D	29	B
10	D	20	A	30	D

A: 6, B: 8, C: 8, D: 8